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Toward Human-Centered Explainability: Natural Language Explanations for Anomaly Detection

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Abstract
This paper proposes a human-centered explainable artificial intelligence pipeline for anomaly detection, designed to generate meaningful, context-aware explanations using local large language models. The proposed pipeline translates model outputs and SHAP-based feature attributions into natural language explanations for cybersecurity alerts generated by an autoencoder within an enterprise network. It incorporates a human-in-the-loop component to ground the explanations in validated expert knowledge, enhancing their interpretability and alignment with human decision-making processes. Using a rubric-driven LLM-as-a-Judge evaluation, we benchmark several large language models and show that as smaller models receive more contextual grounding through human-in-the-loop, their explanatory performance improves significantly, narrowing the gap with larger models while maintaining substantially lower computational demands. Our approach provides targeted, context-aware explanations designed to meet the cognitive and operational needs of security analysts, contributing to more ethical, trustworthy, and resource-efficient AI integration in critical cybersecurity environments.

Keywords Human-centered explainable AI · Natural language explanations · Human-in-the-loop · LLM-as-a-Judge · Anomaly detection

1 Introduction

The increasing integration of artificial intelligence (AI) systems in critical domains such as healthcare, finance, justice, public policy, or cybersecurity has consolidated their role as active agents in decision-making processes with direct consequences for individuals and organizations. In particular, anomaly detection in network traffic flows using machine learning (ML) techniques has become an essential component of modern cyber defense systems (Wang et al., 2021). Both supervised and unsupervised methods enable the identification of anomalous behaviors and the detection of zero-day attacks, whose early discovery is crucial for the protection of sensitive infrastructures.

However, the use of increasingly complex and high-performing models in these contexts introduces a growing risk: the loss of transparency in the decision-making process. Many of these models operate as opaque structures or "black boxes" producing outputs whose underlying logic is unintelligible to end-users (Sovrano & Vitali, 2024). This lack of explainability not only undermines trust in AI-driven decision-making processes but also raises ethical and legal concerns related to its fairness, traceability, and human agency in contexts where AI-driven decisions can have critical implications (Zha et al., 2025).

In response to these challenges, explainable artificial intelligence (XAI) has emerged as an active area of research, aimed at equipping AI systems with mechanisms that facilitate the understanding and auditing of their decisions. Early approaches in XAI focused on providing technical

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explanations targeted at experts, usually through visual or saliency-based methods. Nevertheless, a growing body of research has acknowledged that such explanations, although technically accurate, are often inadequate for non-specialist users or complex operational settings. This realization has driven a paradigm shift toward human-centered XAI (HC-XAI), which advocates for the design of explanations aligned with users' cognitive, informational, and situational needs (Mariotti et al., 2020).

The HC-XAI approach facilitates meaningful interaction between humans and AI systems, emphasizing the need to generate explanations that are not only accurate but also understandable, useful, and context-aware (Mariotti et al., 2020). This alignment is no longer just a design ideal but is becoming a regulatory imperative in high-risk applications (Nannini et al., 2024). The generation of context-aware explanations implies taking into account domain-specific constraints such as data privacy, available computational resources, and users' cognitive load. Moreover, it is widely recognized that no universally valid method exists for producing effective explanations: different contexts require different approaches. In particular, combining textual and visual explanations has been suggested as a way to improve user understanding, and natural language explanations (NLEs) are increasingly seen as a promising path to more transparent, responsible, and trustworthy explanations (Sovrano & Vitali, 2024).

Despite the potential of these techniques, their practical application still faces significant challenges. Producing high-quality NLEs requires integrating outputs from predictive models with XAI techniques and enriching them with contextual information and domain expertise. While many studies employ large language models (LLMs) to generate explanations, their utility decreases without sufficient human alignment and contextualization. Some researchers attempt to address these limitations by adopting larger, more capable LLMs. However, hosting large-scale LLMs locally is often impractical, and systems that employ these models generally rely on cloud-based APIs. This introduces certain regulatory risks, as transmitting potentially sensitive data to third-party servers can conflict with data-protection or data-sovereignty requirements in certain jurisdictions. In addition, explanation quality is often evaluated in a decontextualized manner, without considering how users interpret and act upon these explanations within real-world decision-making workflows.

Motivated by these limitations, the present work proposes an integrated approach aimed at advancing toward truly human-centered XAI. The main contributions of this article are threefold. First, we present a comprehensive pipeline that combines unsupervised anomaly detection, model explainability, and NLE generation with human-in-the-loop (HITL) powered by lightweight, locally hosted LLMs, designed to produce comprehensible and actionable security alerts aligned with regulatory expectations. The same LLMs also serve as a conversational agent, helping analysts in their understanding process through a question-and-answer interface (Abbass et al., 2024). Second, we conduct a rigorous empirical evaluation of various explanation generation strategies, exploring the role of domain expertise through different techniques to enhance contextual precision. Third, we analyze how the interaction between model size and the use of contextual information affects the clarity and adequacy of explanations, demonstrating that smaller yet well-grounded models can offer competitive performance with significantly lower computational overhead.

The remainder of this article is organized as follows. Section 2 provides a comprehensive review of the state of the art in Network Anomaly Detection, XAI, HC-XAI, and the generation of NLEs within the context of anomaly detection. Section 3 introduces the proposed architecture, offering a detailed description of the pipeline and the methodology adopted, including the models evaluated, the dataset used, and the performance metrics considered. In Section 4, we present and analyze the empirical findings, highlighting the comparative effectiveness of the explanation strategies examined. Finally, Section 5 summarizes the main contributions of the study and discusses potential opportunities for future research.

2 Background and Related Work

Making network anomaly detection systems both effective and transparent involves bringing together advancements in different research areas. This section reviews relevant literature in the field of network anomaly detection, then analyzes how XAI is being applied to enhance model's transparency and, finally, the emerging field of NLEs as a crucial step toward HC-XAI.

2.1 Network Anomaly Detection

The adoption of ML techniques for detecting cyberattacks and suspicious activities has emerged as a significant research area within cybersecurity, driven by the continuous evolution of enterprise networks and the increasing complexity of cyber threats. For example, in Hosseinzadeh et al. (2021), authors conduct a comprehensive review of anomaly detection schemes that utilize Support Vector Machines (SVMs), for identifying intrusions. Their work investigates and categorizes various approaches where SVM classifiers are employed, often in conjunction with other ML techniques. Similarly, Keserwani et al. (2021) propose an

approach to anomaly detection in IoT traffic, using a Random Forest classifier to identify anomalies. This model is then evaluated on multiple datasets, to assess its capacity for detecting network attacks. Riyaz and Ganapathy (2020) proposes a deep learning approach for intrusion detection in wireless networks, including a novel feature selection algorithm to feed a Convolutional Neural Network, achieving a 98.8 % attack detection accuracy.

These approaches treat cyberattack detection as a classification problem in a supervised setting, which can limit their ability to adapt to novel or evolving attack patterns (Keserwani et al., 2021). To adapt models to new threats, many works introduce unsupervised anomaly detection methods. For example, in Zoppi et al. (2021), authors conduct an experimental comparison of 17 different unsupervised anomaly detection algorithms across 11 publicly available attack datasets, evaluating their effectiveness. Other popular methods rely on deep learning, with autoencoders being a leading example. Ortega-Fernandez et al. (2024) proposes an unsupervised Intrusion Detection System (IDS) for industrial control systems which utilizes a deep autoencoder for anomaly detection. The authors demonstrate that this technique can achieve high detection rates and low false alarms, outperforming both supervised state-of-the-art methods and unsupervised baselines. Similarly, Torabi et al. (2023) presents an autoencoder model for detecting anomalies in cloud networks. Authors introduce a modification in how the reconstruction error is calculated for anomaly detection that was able to improve accuracy while being an efficient way to identify and classify network attacks. Building upon the autoencoder architecture, Narmadha and Balaji (2025) combines this architecture with an LSTM network. With this integration, the autoencoder is trained to learn normal network patterns and reduce data dimensionality, while the LSTM captures complex sequential dependencies in network data.

While deep learning solutions can excel at anomaly detection, they usually lack transparent decision-making processes due to the use of neural network based ML models, which constrains the adoption of these solutions in critical decision-making scenarios (Ribeiro et al., 2016). In cybersecurity contexts, especially in the domain of intrusion detection, understanding why an anomaly was identified as such is crucial for several reasons: first, it provides interpretability, allowing security analysts to understand the characteristics that are indicative of potential threats, enhancing confidence in the system's decisions. Second, explanations can facilitate actionable insights, enabling targeted investigation, facilitating the implementation of mitigation strategies, and reducing the mean time to resolve a particular cybersecurity incident.

2.2 Explainable Artificial Intelligence

The increasing complexity and "black box" nature of modern AI systems have led to a growing need for transparency and understanding in their decision-making processes. This has given rise to the field of XAI, which aims to provide insights into why AI models make certain predictions and how they arrive at their results (Somani et al., 2023). Current state-of-the-art in XAI can broadly be categorized into model-agnostic and model-specific techniques.

Model-agnostic techniques treat the ML model as a "black box", focusing typically on explaining individual predictions based on input-output relationships. Surrogate model methods exemplify this by employing simpler, human-comprehensible models, such as decision trees or linear models, to explain predictions from opaque systems. Following this approach, LIME (Ribeiro et al., 2016) generates explanations by training a local surrogate model on perturbed data from the original instance. Then, the model approximates the behavior of the original model in that region. Similarly, the SHapley Additive exPlanations (SHAP) method (Lundberg & Lee, 2017) treats features as players in a cooperative game, using Shapley values to measure their contribution to the model's predictions. Despite challenges like choosing baselines and the algorithm's computational efficiency, SHAP is widely used for its strong theoretical foundation.

In contrast to model-agnostic approaches, model-specific methods leverage the internal structure of the ML model to generate explanations. Decomposition-based techniques, such as Layer-wise Relevance Propagation (LRP, Binder et al., 2016) and Deep Taylor Decomposition (DTD, Montavon et al., 2017), break down the relevance score into linear contributions from the input. These methods propagate relevance scores through the network layers to decompose the prediction back to input features or contributions. Other methods in this category include gradient-based explanations, with Integrated Gradients (IG, Sundararajan et al., 2017) serving as their most representative technique. The fundamental mechanism of IG, and its variants, is to accumulate the gradients obtained as the input is interpolated between a reference point and the actual input.

The use of XAI in the cybersecurity domain has been explored in Siganos et al. (2023). Authors incorporated SHAP within an IDS tailored to detect and classify cyberattacks in IoT environments, providing both local and global feature attributions, which are then presented visually. Similarly, Namrita Gummadi et al. (2024) introduce an XAI framework for anomaly detection in IoT environments. This framework integrates various XAI techniques, highlighting key features and their classification thresholds. Kobayashi et al. (2023) extend the LRP framework with DTD to interpret

anomalies generated by an autoencoder, using tailored rules for the reconstruction loss function. However, Sixt et al. (2020) demonstrated that the propagation rules used in these methods often lead to explanations that are independent of the model's later decision-making layers. Sipple and Youssef (2022) introduce a new clustering-based approach of normal baselines from training data to generate contrastive explanations, and applying IG for feature attribution. Aiming to improve the interpretability of deep autoencoders in anomaly detection, Antwarg et al. (2021) proposes a method based on SHAP to explain high reconstruction error features, addressing the challenge of explaining co-occurring anomalous features. The evaluation showed that this method proved to be more robust and effective than LIME.

Despite significant advancements in XAI techniques, the development of human-centric explanation methods remains in its early stages. This gap often results in solutions that, while technically robust and effective, are not easily understandable to non-expert users, failing to fulfill the core objective of XAI systems: provide actionable and understandable insights (Miller, 2019).

2.3 Natural Language Explanations in Cybersecurity

Researchers have explored various ways to approach human-centered XAI, but there is no single, universally accepted method (Sovrano & Vitali, 2024). However, in Kayser et al. (2024), authors compared visual, text-based, and combined explanations in a medical environment for both expert and non-expert users. They evaluated how different explanation types influence users. Similarly, in Szymanski et al. (2021), authors found that users often prefer visual explanations over text-based, but they frequently misunderstand them. Both studies agreed that the combination of textual and visual explanations is the best approach, with text being the most promising solution (Mariotti et al., 2020).

However, generating dynamic, text-based explanations for XAI which align with specific user needs or contexts outside laboratory environments can be challenging. This process involves leveraging model outputs, interpretability techniques, and contextual adaptations to produce explanations. One of the first studies to explore the field of NLE with LLMs for XAI was Burton et al. (2023). In this work, authors present a dataset comprised of expert-written textual narratives that explain XAI outputs. This dataset was used to fine-tune two LLMs, BART (Lewis et al., 2020a) and T5 (Raffel et al., 2020). The quality of the generated text was assessed using automatic metrics such as BLEU (Papineni et al., 2002) and ROUGE (Lin, 2004), and a manual error analysis on 30 test records of the narratives generated by BART. Another example of a combination of visual explanations and dynamic text explanations is HuntGPT (Ali & Kostakos, 2023) which uses a two-step strategy integrating SHAP and LIME with a Random Forest model for anomaly detection and an NLE component via a GPT-3.5 Turbo to provide interpretable insights into the detected network anomalies. This integration, in particular the conversational interface, can significantly improve user understanding, enabling a more robust and user-friendly mechanism to generate explainable and actionable XAI solutions in cybersecurity operations. Similarly, Baral et al. (2024) employ the same two-step strategy: first, using traditional XAI to reveal influential features, and second, integration of an LLM to translate the XAI outputs into human-understandable explanations. The authors conducted a comparative analysis of the responses from Gemini and OpenAI to a specific cybersecurity prompt regarding a DoS TCP Flood attack prediction from a Random Forest model. Finally, Lim et al. (2025) proposes a framework designed to enhance both detection accuracy and explainability. They leverage LLMs to translate SHAP and LIME outputs into NLEs, targeting a use case of email phishing detection.

However, while these studies demonstrate the potential of LLMs to translate technical data into accessible narratives, they also highlight a critical dependency on the pre-trained knowledge of these general-purpose models (Ali & Kostakos, 2023). The cybersecurity domain is highly specialized and dynamic, with context-specific threats, organizational policies, and network architectures that are not present in public training data. A promising direction for overcoming this limitation is the adoption of HITL, where AI integrates expert feedback to enhance model's accuracy, adaptability, and interpretability (Yang et al., 2023) to ensure the factual domain accuracy of the LLM's output, and adjust the language and recommendations to specific environments, enabling the LLM-based system to evolve from generic translators to specialized assistants.

Multiple approaches can be followed to integrate HITL into an explanation pipeline (Patil & Gudivada, 2024). One method is the domain adaptation, where fine-tuning the model on high-quality, domain-specific data allows the LLM to extend its knowledge and align with the nuances, terminology, and reasoning patterns of a particular domain (Bommasani et al., 2022). Another approach involves implementing feedback loops, which enable experts to interact with the LLM and provide feedback on its outputs, typically using reinforcement learning to adjust model behavior accordingly (Ouyang et al., 2022). An alternative strategy is contextual retrieval from human knowledge, which uses retrieval-augmented generation (RAG, Lewis et al., 2020b) to bring in similar expert-written explanations, combining them with prompt engineering so the LLM can reason using validated human expertise (Arslan et al., 2024).

Each method presents different trade-offs. Domain adaptation deeply aligns the model with specific expert knowledge but requires extensive training data and can become outdated. Feedback loops can iteratively refine outputs to build user trust, yet their success hinges on a well-calibrated reward function, which can be opaque. Finally, contextual retrieval offers a flexible, up-to-date grounding mechanism but is constrained by the quality of the retrieval process and the potential of inheriting expert biases. The selection of an appropriate method, therefore, depends on the specific requirements for data availability, model adaptability, and runtime performance.

Finally, even though recent advancements, a major challenge in this field is the evaluation of the quality of the model's outputs. Traditional automated methods, like BLEU (Papineni et al., 2002) and ROUGE (Lin, 2004), are not sufficient for assessing complex explanations, since these metrics cannot accurately measure the alignment of the outputs with human preferences or domain's requirements. While in this context human evaluation is seen as the gold standard, it's often too expensive, time-consuming, and inconsistent for large-scale use (Deriu et al., 2021).

To overcome these issues, the LLM-as-a-Judge paradigm emerged as an effective alternative (Zheng et al., 2023). This paradigm leverages LLMs as evaluators, demonstrating that with clear guidelines, their judgments on qualities like alignment or coherence align remarkably well with human experts. However, this method is not without limitations: AI judges can have biases, such as preferring longer or first-seen answers, or generate hallucinations (Li et al., 2024). While various mitigation strategies exist, a particularly effective approach (especially for domain-specific tasks), is the implementation of a structured rubric (Peng et al., 2023; Zheng et al., 2023). By employing a structured rubric, the judge's assessment is guided by specific criteria, leading to an evaluation that is more transparent, replicable, and aligned with domain requirements (Li et al., 2025).

Despite growing interest in HC-XAI for cybersecurity, existing approaches fail to offer human-centered explanations. Many explainability methods either stop at highlighting anomalous features or rely heavily on visual outputs without evaluating their practical utility for security analysts. Critically, they lack robust evaluation frameworks and fail to incorporate domain expertise into the explanation process. Our work addresses these limitations by introducing a novel evaluation framework, focused on assessing explanations against criteria essential to an analyst, such as threat awareness. Additionally, we present a pipeline that generates NLEs, enriched with domain-specific knowledge, and delivers them via an interactive chatbot interface designed to facilitate analysts' comprehension and engagement with security alerts.

3 Methodology

This section introduces the proposed methodology for HC-XAI in the cybersecurity domain, which addresses key shortcomings in current approaches by integrating unsupervised anomaly detection, contextualized explanation generation, and HITL feedback to support analyst decision-making. Figure 1 depicts the proposed HC-XAI workflow, designed to translate predictions of a deep learning anomaly detection model into actionable NLEs for Security Operations Center (SOC) analysts. The adopted methodology is structured into five distinct phases: (1) Data Analysis & Preprocessing (2) Anomaly Detection, (3) SHAP integration, (4) Human-Centered Explanations, and (5) NLE alignment. A complete example of the proposed explainable detection pipeline is available on GitHub.[1]

3.1 Data Analysis & Preprocessing

The evaluation of the proposed approach is performed using the CIC-IDS-2018 dataset (Sharafaldin et al., 2018), a complex benchmark dataset for intrusion detection research, available from the Canadian Institute for Cybersecurity. This dataset was selected due to its realistic representation of modern IT environments, encompassing a wide range of benign and malicious network activities, including sophisticated cyberattacks such as infiltration, brute-force, distributed denial-of-service (DDoS), botnets, among others. The experimental setup of the dataset involved an attacker infrastructure of 50 machines targeting a simulated organizational network, structured into five departments, comprising 420 computers and 30 servers. This dataset is composed of bidirectional network flows, which were extracted from captured network traffic using CICFlowMeter-V3. Table 1 presents the traffic capture days for each traffic type along with the corresponding number of flows collected. Consistent with many network-based intrusion detection datasets, benign flows significantly outnumber anomalous ones.

Figure 2 illustrates a t-SNE visualization for understanding the characteristics of the CIC-CSE-IDS-2018 dataset and, consequently, the inherent challenges in anomaly detection. Contrary to an idealized expectation where attack traffic would manifest as distinct, well-separated clusters, the visualization clearly reveals a substantial degree of overlapping between benign traffic and some attack categories, such as infiltration and benign traffic. Such similarity implies that these attacks in the CIC-CSE-IDS-2018 dataset are not superficial outliers but rather exhibit characteristics deeply integrated with normal traffic distributions, which

[1] https://github.com/Gradiant/Natural-Language-Explanations-4-AD

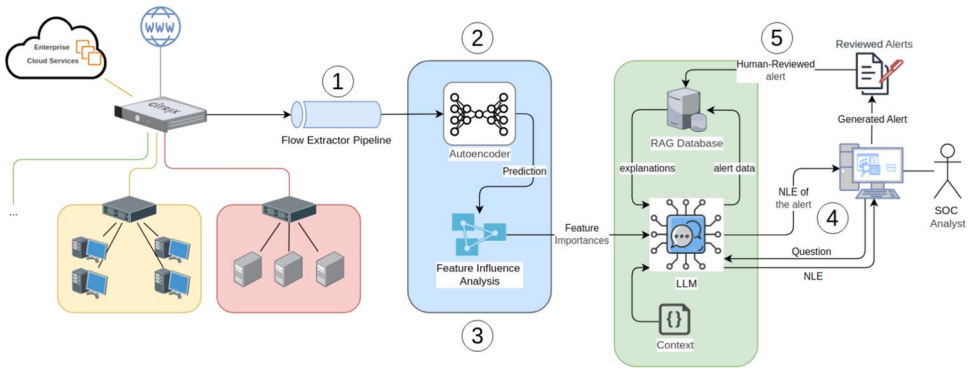


Fig. 1 The proposed HC-XAI workflow, showing the pipeline from network flow extraction to the generation and review of human-understandable security alerts

Table 1 Distribution of classes in the CIC-IDS2018 dataset, detailing the number of network flows and capture days for each benign and attack type. The table highlights the significant class imbalance inherent in the data

Type	Number of flows	Capture days
Benign	13484708	10
DDOS attack-HOIC	686012	1 (21/02/2018)
DDoS attacks-LOIC-HTTP	576191	1 (20/02/2018)
DoS attacks-Hulk	461912	1 (16/02/2018)
Bot	286191	1 (02/03/2018)
FTP-BruteForce	193360	1 (14/02/2018)
SSH-Bruteforce	187589	1 (14/02/2018)
Infiltration	161934	2 (01/03-28/02/2018)
DoS attacks-SlowHTTPTest	139890	1 (16/02/2018)
DoS attacks-GoldenEye	41508	1 (15/02/2018)
DoS attacks-Slowloris	10990	1 (15/02/2018)
DDOS attack-LOIC-UDP	1730	2 (20-21/02/2018)
Brute Force -Web	611	2 (22-23/02/2018)
Brute Force -XSS	230	2 (22-23/02/2018)
SQL Injection	87	2 (22-23/02/2018)

makes them harder to detect and reflects the complex nature of today's cyberattacks.

Prior to model training and evaluation, the raw data went through several preprocessing steps. First, the different capture days were aggregated and merged to form a unified dataset, except for 20/02, which showed certain capture inconsistencies, followed by a data cleaning step. To address the presence of outliers and facilitate convergence during training, a two-step scaling approach was adopted. Initially, Robust Scaling removes the median and scales the data based on the interquartile range to minimize the influence of outliers. Next, Min-Max Scaling normalizes the features to a consistent range between 0 and 1 to help with the optimization process during training. Regarding feature selection, no features were removed from the dataset due to inter-feature correlations. This decision was deliberate, as some unsupervised models, such as autoencoders, are particularly good at learning the relationships that exist among input variables.

3.2 Anomaly Detection

Following the preprocessing of the dataset, this phase focuses on the implementation and training of the anomaly detection model. For this purpose, an unsupervised deep learning model, specifically a deep autoencoder (Rumelhart et al., 1986), was selected. Autoencoders are particularly well-suited for anomaly detection tasks due to their ability to learn complex, non-linear representations of normal behavior from unlabeled data. By training the autoencoder exclusively on benign traffic, it learns to reconstruct normal patterns. Consequently, when presented with anomalous or malicious traffic, which deviates significantly from these learned patterns, the model will exhibit a higher reconstruction error (the discrepancy between the original input and the autoencoder's reconstruction).

An instance can be classified as an anomaly if its reconstruction error exceeds a predefined threshold. This threshold can be determined by analyzing the distribution of reconstruction errors on a validation set of unseen benign data. Therefore, each defined experiment involves four specific subsets of benign instances: training (60 %), used to train the autoencoder; validation (10 %), used for model validation during training; threshold (10 %), reserved for determining the anomaly detection threshold; test (20 %): used for final performance assessment.

To determine the optimal configuration of the model, a hyperparameter tuning was performed. Various deep autoencoder neural network architectures, with different number of hidden layers and neurons at each layer, and other parameters like dropout and regularization were evaluated. The selected deep autoencoder architecture consisted of 6 hidden layers for both the encoder and the decoder plus the latent space. Each hyperparameter training was performed for 100 epochs (with early stopping) using the Mean Squared Error (MSE) as the loss function. The final model used a learning rate of 0.01, a batch size of 256, a dropout rate of 0.01, the ReLU activation function, and an L2 regularization value of 0.001.

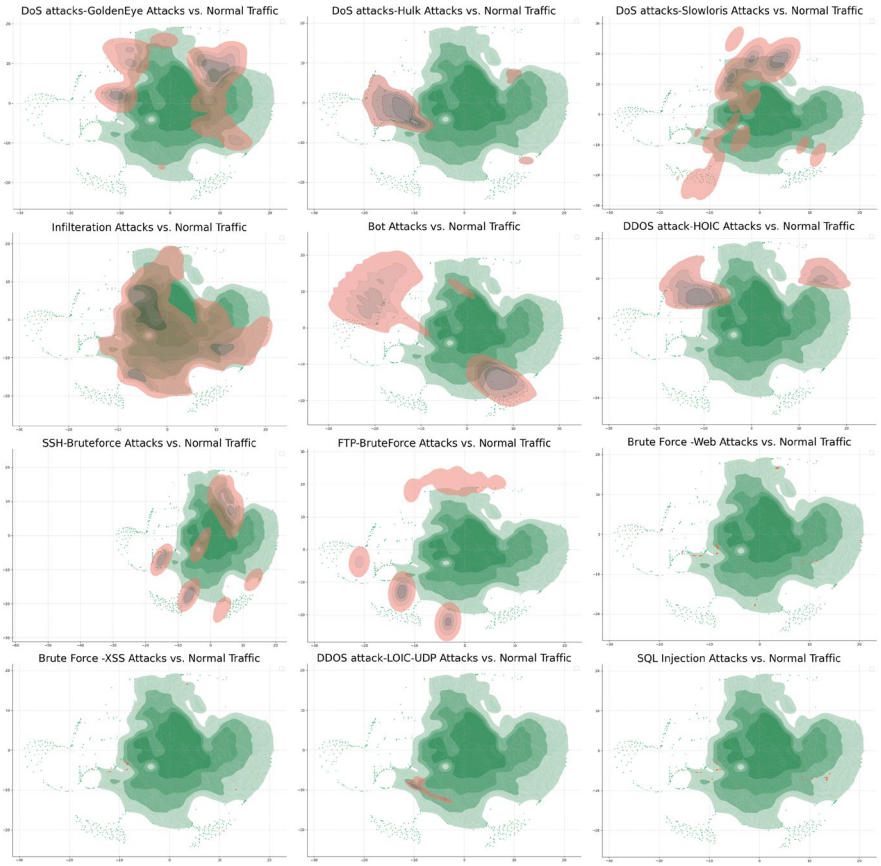


Fig. 2 t-SNE visualization of attack categories in CIC-IDS-2018. Green clusters represent the aggregated benign traffic from all days, while the red clusters represent the distinct attack types within the dataset

The performance of the deep autoencoder was assessed using a 10-fold cross-validation methodology. As the model's objective is to minimize the reconstruction error, a threshold was established to distinguish anomalies from benign instances. For each fold, this threshold was defined as the 95th percentile of the reconstruction errors from the benign data, enabling performance evaluation with the following binary classification metrics: accuracy, precision, recall, F1-Score, Area Under the ROC Curve (ROC-AUC Score) and Area Under the Precision-Recall Curve (PR-AUC Score). A limitation of these metrics is their inability to differentiate performance across specific attack types, since all malicious traffic is treated as a single anomaly class. While a multi-class evaluation would provide a more granular assessment, it is incompatible with the autoencoder's architecture. To address this limitation, detection rates for each individual attack are also reported.

3.3 SHAP Integration

While autoencoders have a remarkable performance at unsupervised anomaly detection, they operate as "black box" models. However, features exhibiting high reconstruction error are, in a primary sense, self-explanatory because the magnitude of their error directly indicates a significant discrepancy. Nevertheless, prior research (Antwarg et al., 2021) has shown that relying solely on reconstruction errors or raw feature values alone often leads to incomplete or misleading interpretations of why the autoencoder failed to reconstruct specific inputs correctly.

Autoencoders, by their nature, learn complex, often non-linear, interdependencies, and correlations among input features. Consequently, the poor reconstruction of a particular input feature is often not just due to an anomaly in that specific feature. Instead, its value may be anomalous in the context of other co-occurring features. While these learned correlations are fundamental to the effectiveness of the autoencoder, they also mean that the reconstruction of one feature is influenced by many others. Antwarg et al. (2021) demonstrate that incorporating SHAP-based feature attributions alongside reconstruction errors significantly improves explanation quality by clarifying these interdependencies, something that raw feature values or reconstruction errors alone cannot provide.

Building on this, this work modified the SHAP framework (Lundberg & Lee, 2017) to adapt to the characteristics of the autoencoder model, following the approach of Antwarg et al. (2021). For any instance identified as an anomaly, the process begins by isolating the top N input features that exhibit the highest reconstruction error computed as the

MSE between the original input and its reconstruction. For each feature, its reconstructed value becomes the specific prediction that SHAP aims to explain, with the autoencoder as the prediction function. To this end, KernelSHAP (Lundberg & Lee, 2017) is used to estimate the SHAP values by observing how its reconstructed value changes as it samples groups of input features against a baseline expectation. This SHAP variant combines the principles of linear LIME (Ribeiro et al., 2016) with Shapley values to construct a local explanation model. This model takes the form of a weighted linear regression, built using a background dataset and samples from the possible coalitions of input features. A crucial design choice in this variant is the definition of the background data, which represents the reference distribution of normal inputs. This baseline captures the diversity of benign behavior, constructed from the subset of clean, non-anomalous training data. This set is then summarized using K-Means clustering (Macqueen, 1967), reducing the background set from 13 million samples to 100 centroids. This approach maintains computational efficiency while providing a balanced and representative baseline for KernelSHAP. The full procedure is detailed in Algorithm 1.

Algorithm 1 Integration of SHAP with autoencoders (Antwarg et al., 2021).

Input: x: original anomalous instance, x': reconstructed instance, B: K-Means of baseline, ae: autoencoder model, N: top-error features to explain.
Output: SHAP values for each high-error feature

procedure EXPLAIN(x, x', B, ae, N)
  $topErrors \leftarrow$ MSEPERFEATURE(x, x')
  $explainer \leftarrow$ KERNELEXPLAINER($ae.predict, B$)
  for each $feature \in topErrors$ **do**
    $shapValues_{feature} \leftarrow$ EXPLAINER.SHAPVALUES($x'_{feature}$)
    $shapExplanations_{feature} \leftarrow shapValues_{feature}$
  end for
  return $shapExplanations$
end procedure

To exemplify the output of Algorithm 1, Fig. 3 presents the proposed visualization of the SHAP-based method for $N = 3$. For each high-error feature, the corresponding subplot depicts the most influential features and their impact on the feature's error. Features shown in red increase the reconstruction error, contributing to the anomaly, while those in blue have an offsetting effect.

3.4 Human Centered Explanations

Although feature-level attributions from XAI techniques like SHAP improve model transparency, they are often too technical for immediate interpretation. Converting these outputs into NLEs makes them more comprehensible and better aligned with domain conventions. LLMs provide an effective mechanism for automatically generating such NLEs. However, despite their remarkable capabilities, LLMs are not without limitations (Ji et al., 2023; Ye & Durrett, 2022). They may produce explanations that are overly generic, contextually inaccurate, or misaligned with the actual behavior of the underlying predictive model. These issues can arise due to the reliance of LLMs on training data that may not fully capture the complexities of specific domains, such as cybersecurity (Ali & Kostakos, 2023; Hasanov et al., 2024), or because of challenges in interpreting structured SHAP values. To mitigate these challenges and ensure the reliability of generated explanations, HITL (Yang et al., 2023) techniques were recently introduced to incorporate expert knowledge into the AI explanation process.

Considering the characteristics of the methods reviewed in Section 3, this work adopts a contextual retrieval approach using RAG (Lewis et al., 2020b), which combines the strengths of information retrieval and LLMs, enabling more accurate and context-aware responses by incorporating external knowledge into the response generation process. In this context the HITL aspect refers to the process of continually improving the system's explanations with expert knowledge provided by the RAG component. The human expert comes into the loop by expanding this knowledge base in the RAG component: when a cybersecurity analyst determines that an LLM-generated explanation is incomplete or could be improved, he/she can write a more precise, expert-level explanation. This new, human-written

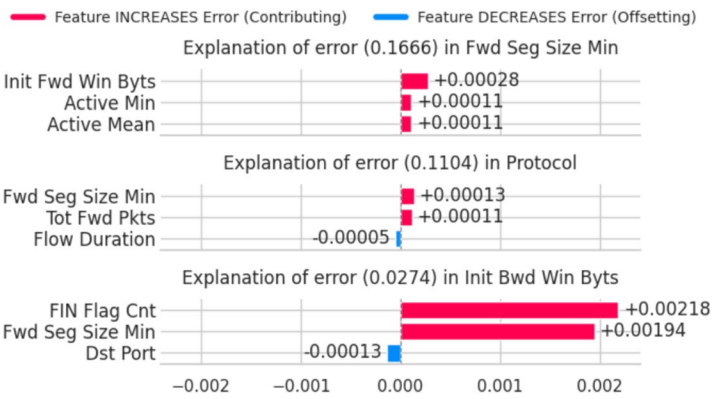


Fig. 3 SHAP-based proposed visualization for a sample anomaly. The visualization shows the top three features with the highest reconstruction error and identifies the features that most influenced each error

explanation is then fed back into the RAG database (represented in the step 5 of Fig. 1). These refined explanations will be retrieved by the LLM in future analysis, helping the model to produce better explanations. Over time, the system gradually produces clearer and more accurate explanations, since it can rely on an expanding collection of well-written expert contributions. This approach offers substantial flexibility and adaptability for organizations, as the knowledge base can be updated frequently or at longer intervals according to operational needs. For instance, zero-day incidents can be integrated directly into the knowledge base, enabling the system to handle emerging threats without model (LLM) retraining. This database is then used by the LLM to ground the NLE process by retrieving the top-k most similar pairs of *(alert, explanation)* written or validated by domain experts, using the cosine similarity metric.

The experiments in this paper use a value of $k = 3$. This value was chosen to ensure explanation diversity and prevent the model from basing its explanation in any single instance. Using fewer examples ($k < 3$) can lead to biased and less diverse explanations. On the other hand, a higher k would exponentially increase processing time and could also reduce explanation quality. Because the models used in this work have a limited context window, they may struggle to maintain coherent attention across a bigger input, potentially overemphasizing the most recent tokens and producing explanations that are less consistent or "overfit" to the latest context. Therefore, the authors consider $k = 3$ to be an efficient and balanced choice for generating context-aware explanations.

To assess the effectiveness of the proposed HITL approach, the evaluation encompasses both the retrieval and generative components of the HC-XAI system. The retrieval accuracy is evaluated on the CIC-IDS-2018 dataset, measuring how well the system can retrieve one or more relevant explanations that are associated with the same attack type as the current alert. The experimental design of the generative components evaluates a diverse set of foundational models, including Mistral-7B-Instruct-v0.3 (Mistral-AI, 2024), Qwen1.5-4B-Chat (Bai et al., 2024b), Gemma-2-9b-it (Google, 2024), Llama-3.1-8B-Instruct (Meta, 2024), and Qwen2.5-32B-Instruct (Bai et al., 2024a), across three NLE strategies. The notation "[Number]B" in a model's name specifies its size in billions of parameters. For example, Gemma-2-9b-it is a model with 9 billion parameters. Parameters are the internal weights that the model learns during training, and the number of parameters generally correlates with the model's capacity to understand and generate complex language patterns.

Each NLE strategy represents a different level of contextual guidance to determine the most effective method for generating high-quality NLEs. The first, a zero-shot prompt, served as a baseline by providing the LLM with minimal instruction and relying primarily on its pre-trained knowledge to interpret the raw XAI outputs. The second, a contextualized prompt with a template, introduced a domain-adapted approach that incorporated cybersecurity-specific terminology and enforced a structured format. The third and most advanced strategy used the RAG approach, which dynamically grounds the LLM's response by retrieving relevant, expert-curated explanations from an external knowledge base, ensuring the final NLE is aligned with validated human expertise. Details and descriptions of the prompts used in each NLE strategy, along with the corresponding evaluation dataset, are publicly available at GitHub.[2]

To evaluate the effectiveness of the different NLE strategies described, the methodology LLM-as-a-Judge (Zheng et al., 2023) was chosen. The evaluation process was guided by a structured rubric, available in Table 2, and encompasses five key dimensions: Content Accuracy, Clarity and Coherence, Relevance and Focus, Conciseness, and Threat Awareness. This rubric was embedded directly within the evaluation prompt. Candidate explanations were evaluated in the context of AI-generated alerts corresponding to specific cybersecurity attack types (e.g., Infiltration, DoS).

The evaluation data used to assess the LLMs in the different NLEs strategies was constructed through a sampling approach, ensuring equal coverage of different attack types, while maintaining computational feasibility. Attack instances in the CIC-IDS-2018 dataset were first identified by the trained autoencoder. From the pool of detected anomalies, 30 samples for each attack type were randomly selected to create a balanced evaluation set. For each selected sample, the SHAP-based method in Algorithm 1 was applied. These outputs were then formatted and fed as input to each of the evaluated LLMs using the three different prompting strategies. To construct the *(alert, explanation)* pairs used in the HITL strategy, a subset of CIC-IDS-2018 alerts, available on GitHub,[3] that were not included in the evaluation set was selected. From this subset, ten alerts were sampled from each attack category and associated with a human explanation of the alert derived from autoencoder's XAI outputs and the official CIC-IDS-2018 documentation. The resulting NLEs were collected and stored for subsequent evaluation using the LLM-as-a-Judge framework.

[2] https://github.com/Gradiant/Natural-Language-Explanations-4-AD

[3] https://github.com/Gradiant/Natural-Language-Explanations-4-AD

Table 2 Evaluation rubric for LLM-generated cybersecurity explanations

Criterion	Score range	Description
Content accuracy	9–10	Fully aligns with SHAP data, accurately interprets anomalies, and connects them to plausible threat scenarios.
	5–8	Partially aligns with SHAP data; contains minor inaccuracies or incomplete interpretations.
	0–4	Fails to align with SHAP data and provides inaccurate or meaningless interpretations.
Clarity and coherence	9–10	Well-structured, free of spelling errors, and avoids unnecessary jargon.
	5–8	Lacks clear structure or contains minor spelling errors.
	0–4	Poorly structured, includes errors, and uses excessive or unclear jargon.
Relevance and focus	9–10	Contains only information directly related to the alert's features or threat scenarios.
	5–8	Includes some irrelevant details or loosely connected information.
	0–4	Contains mostly irrelevant or unrelated information.
Conciseness	9–10	Addresses all SHAP values effectively and remains concise.
	5–8	Addresses some SHAP values; may be verbose or omit key points.
	0–4	Omits most SHAP values or is overly wordy and unfocused.
Threat awareness	9–10	Threat scenarios are plausible, creative, and grounded in cybersecurity knowledge.
	5–8	Scenarios are somewhat plausible but may lack depth or creativity.
	0–4	Scenarios are implausible, generic, or unsupported by relevant cybersecurity context.

4 Results and Discussion

This section presents the experimental results of the proposed anomaly detection pipeline to generate NLEs augmented with domain knowledge. The results are organized in two main parts: first, the performance of the autoencoder anomaly detection model is evaluated; second, the effectiveness of the generated NLEs is assessed using a human-centered evaluation protocol based on the LLM-as-a-Judge evaluation.

4.1 Autoencoder Model Performance

The performance of the deep autoencoder for unsupervised anomaly detection was evaluated using a set of metrics across the CIC-IDS-2018 dataset. The evaluation employed 10-fold cross-validation to ensure robust and reliable performance estimates, with results presented as mean values and standard deviations across all folds. Table 3 presents the evaluation results for the autoencoder model across all attack types. The model demonstrates strong, state of the art performance for the dataset (Verkerken et al., 2021), with an average ROC-AUC of 91.14 %, across all folds.

Table 3 Overall performance metrics of the deep autoencoder across all attack types. Values represent mean ± standard deviation over 10-fold cross-validation. Inference time is reported per instance

Metric	Mean ± Std
Accuracy (%)	89.12 ± 1.16
Precision (%)	86.22 ± 0.66
Recall (%)	75.10 ± 3.87
F1-Score (%)	80.24 ± 2.47
ROC-AUC (%)	91.14 ± 0.73
PR-AUC (%)	86.27 ± 1.21
Inference time (μs)	2794 ± 89

In real-world network traffic and in datasets like CIC-IDS-2018, attacks are not uniformly distributed. The dataset contains a massive number of Benign samples, but among the attacks, there is also a severe imbalance, as shown in Table 1. This imbalance can create a misleading sense of high performance, as a model may achieve a near-perfect score simply by detecting the most common threats while completely failing to identify under-represented attacks like SQL Injection. Relying solely on binary classification metrics would be misleading, as this approach ignores the substantial differences between attack types, represented in Fig. 2.

To provide a complete evaluation, Table 4 presents mean and standard deviation reconstruction error and detection rate of every attack over the 10 folds, using as evaluation threshold the 95th percentile, as explained on Section 3.2. As expected, the detection performance varies significantly across different attack types, with rates ranging from 100 % for some categories to 11 % for others. This highlights the inherent difficulty in identifying complex attacks like Infiltration, Bot and Web attacks, which, as illustrated in Fig. 2, exhibit very close patterns to normal traffic distributions. Furthermore, DDoS-HOIC and Bot attacks showed a high standard deviation, indicating their detection rates are particularly sensitive to the threshold value. This variability highlights a broader issue: certain attack patterns may not be consistently visible in network traffic features, making them inherently harder to detect through flow-based analysis alone, regardless of just threshold tuning.

Nevertheless, the autoencoder demonstrated a strong overall performance, validating its effectiveness in detecting a diverse range of threats and confirming its suitability for providing reliable inputs to the subsequent HC-XAI components.

Table 4 Detailed performance metrics for individual attack types (mean and standard deviation), showing reconstruction errors and detection accuracy

Attack type	Reconstruction error	Detection accuracy
Benign	$(4.85 \pm 0.99) \times 10^{-6}$	94.99 ± 0.05
Slowloris	$(1.60 \pm 0.51) \times 10^{-3}$	89.62 ± 10.69
DDoS-HOIC	$(7.68 \pm 20.86) \times 10^{-4}$	54.21 ± 37.4
SQL Injection	$(1.49 \pm 0.72) \times 10^{-5}$	31.26 ± 11.47
Brute Force-Web	$(1.65 \pm 0.72) \times 10^{-5}$	42.98 ± 6.22
Bot	$(1.33 \pm 1.53) \times 10^{-5}$	30.60 ± 24.83
DDoS-LOIC	1.83 ± 0.03	100.00 ± 0.00
Infiltration	$(1.26 \pm 0.37) \times 10^{-5}$	11.39 ± 0.91
DoS-GoldenEye	$(5.46 \pm 2.21) \times 10^{-5}$	99.74 ± 0.61
Brute Force-XSS	$(2.69 \pm 1.54) \times 10^{-5}$	49.26 ± 0.64
FTP-BruteForce	$(9.30 \pm 5.82) \times 10^{-4}$	99.95 ± 0.16
DoS-SlowHTTPTest	$(5.80 \pm 4.15) \times 10^{-4}$	99.90 ± 0.32
DoS-Hulk	$(6.47 \pm 4.54) \times 10^{-5}$	99.99 ± 0.00
SSH-BruteForce	$(2.12 \pm 1.29) \times 10^{-4}$	99.99 ± 0.00

4.2 Evaluation of Natural Language Explanations

To evaluate the proposed contextual retrieval system, this assessment measures the performance of its two fundamental components: the RAG and the NLE generation. All experiments were conducted in a controlled server environment, utilizing an NVIDIA H100 GPU to ensure consistent and reproducible hardware performance. Furthermore, to optimize performance and ensure viability in real-world scenarios, models were loaded using 4-bit double quantization (Dettmers et al., 2023), a technique that significantly reduces their memory footprint.

To assess the performance of the retrieval component to recover similar pairs of *(alert, explanation)*, the retrieval accuracy on the CIC-IDS-2018 dataset is presented. This retrieval accuracy can be defined as the ability to retrieve one or more relevant explanations that correspond to the same attack type as the current alert. This metric reflects how effectively the retrieval mechanism captures similar instances to support accurate and meaningful explanations. Results detailed in Fig. 4 shows a high retrieval accuracy across most attacks; however, it was significantly lower for the Infiltration category, a direct consequence of the inherently heterogeneous and subtle nature of these attacks.

To evaluate the quality of the generated explanations, the LLM-as-a-Judge paradigm was employed, using GPT-4 as evaluator. Each explanation was scored against a comprehensive rubric, detailed in Table 2, which was designed to reflect the operational needs of a SOC analyst.

The results from the evaluation of the zero-shot prompt, the contextualized prompt, and the contextual retrieval from human knowledge are presented in Table 5. To ensure a robust evaluation across diverse threat scenarios, the performance on each rubric dimension is reported (mean and standard deviation) from a set of 30 randomly sampled instances from each attack type. To quantify the resource demands, Fig. 5 shows the peak GPU memory consumption during the NLE generation for each evaluated model.

The first experiment, the simple prompt, produced explanations with limited value: while dimensions like Clarity and Conciseness achieved moderate scores, the critical metrics for the domain, Content Accuracy and Threat Awareness, were consistently low, averaging around 4-5 points. However, by introducing relevant context and guidance on the prompt, the models' outputs became significantly more aligned with human expectations. Clarity, Relevance,

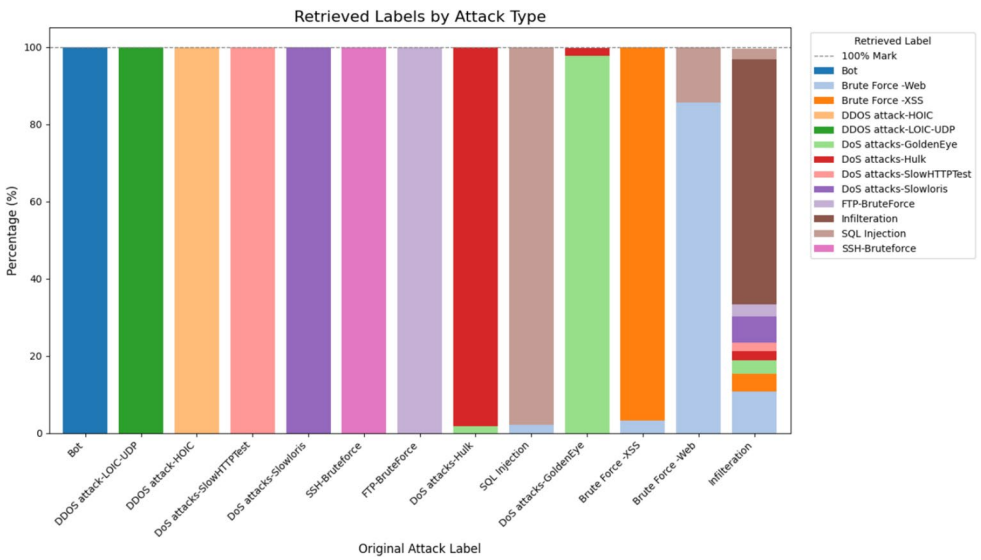


Fig. 4 Composition of retrieved similar labels for each original attack type for $k = 3$. Each bar represents the original label, subdivided by the proportion of predicted labels assigned to it

Table 5 Combined performance metrics for each model across three different prompt strategies. Values are reported as mean ± standard deviation across all alerts

Model	Time (sec.)	Content	Clarity	Relevance	Conciseness	Threat A.
Zero-shot strategy						
Mistral-7B	6.99 ± 1.33	5.05 ± 1.20	6.95 ± 0.96	5.82 ± 1.39	6.55 ± 1.13	5.32 ± 1.36
Qwen1.5-4B	9.22 ± 2.35	4.31 ± 0.96	6.90 ± 1.23	4.88 ± 1.09	6.40 ± 1.14	4.39 ± 1.21
Gemma-9B	14.50 ± 2.20	4.92 ± 1.11	8.07 ± 0.91	6.20 ± 1.42	7.60 ± 0.86	5.30 ± 1.40
Llama-8B	7.26 ± 1.24	4.60 ± 1.14	6.32 ± 1.26	5.39 ± 1.37	6.27 ± 1.37	4.56 ± 1.45
Qwen2.5-32B	21.77 ± 3.00	5.88 ± 1.60	7.90 ± 1.23	7.14 ± 1.53	7.65 ± 1.07	6.10 ± 1.47
Contextualized prompt strategy						
Mistral-7B	12.07 ± 3.35	7.52 ± 1.44	9.15 ± 0.56	8.28 ± 1.42	9.02 ± 0.48	6.95 ± 1.67
Qwen1.5-4B	8.50 ± 2.53	5.98 ± 1.72	8.51 ± 0.78	6.57 ± 1.92	7.89 ± 0.98	5.98 ± 1.70
Gemma-9B	23.56 ± 3.21	7.53 ± 1.28	9.29 ± 0.47	8.31 ± 1.37	9.04 ± 0.42	6.70 ± 1.70
Llama-8B	8.01 ± 1.80	7.54 ± 1.35	9.15 ± 0.44	8.09 ± 1.39	8.84 ± 0.57	6.93 ± 1.42
Qwen2.5-32B	25.26 ± 3.34	7.90 ± 1.20	9.47 ± 0.50	8.64 ± 1.26	9.13 ± 0.47	7.12 ± 1.61
Contextual retrieval strategy						
Mistral-7B	48.68 ± 18.39	8.42 ± 1.35	9.14 ± 0.54	8.58 ± 1.17	7.70 ± 0.84	8.25 ± 1.49
Qwen1.5-4B	28.42 ± 17.91	5.53 ± 2.10	7.79 ± 1.31	5.76 ± 2.11	5.83 ± 1.59	5.78 ± 2.06
Gemma-9B	52.22 ± 18.06	8.66 ± 0.89	9.34 ± 0.48	9.08 ± 0.68	8.30 ± 0.53	8.21 ± 1.27
Llama-8B	49.99 ± 12.68	8.21 ± 1.78	9.01 ± 0.71	8.33 ± 1.70	7.03 ± 1.21	8.20 ± 1.54
Qwen2.5-32B	87.88 ± 27.91	9.06 ± 0.87	9.53 ± 0.50	9.38 ± 0.70	8.37 ± 0.61	8.36 ± 1.32

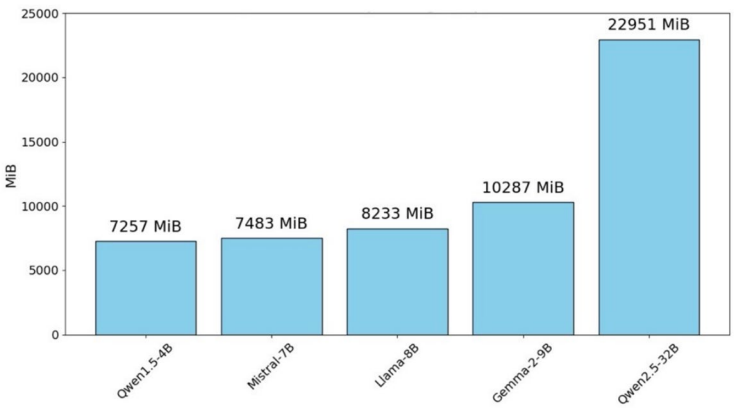


Fig. 5 A bar chart comparing the peak GPU memory allocation (in MiB) for each of the five models evaluated in this study. The memory usage scales with the size of the model, with Qwen2.5-32B requiring the most significant resources

and Conciseness scores rose sharply, with top models like Gemma-9B and Mistral-7B approaching scores of 9 or higher. Other evaluation dimensions also increased substantially. Nevertheless, when HITL was applied, performance in these most critical dimensions, Content Accuracy and Threat Awareness, reached its highest score, with the strongest models scoring in the 8-9 range. These results demonstrate that grounding an LLM in domain-specific knowledge is essential for producing explanations that are not only clear but also technically aligned with the domain and user expectations.

The complete HC-XAI workflow is demonstrated using an Infiltration alert from the CIC-IDS-2018 dataset, with Mistral-7B as the LLM, chosen for its balance between model size size and performance. This walkthrough, available on GitHub,[4] shows the process from raw network features to SHAP-based attribution and finally to natural language explanations. As expected, the effectiveness of the strategies increases with more contextualization. Note that this alert corresponds to the second stage of the attack, which involves an internal IP sweep, a port scan, and service enumeration using Nmap. The zero-shot strategy produces an output that is overly generic, offering only broad threat categories without linking them to the underlying XAI evidence. The contextualized strategy, in contrast, yields a more structured explanation that demonstrates a clearer understanding of the alert, for example, recognizing that the combination of anomalous "Fwd Seg Size Min" and the presence of special TCP flags could indicate port scanning activity. The HITL strategy goes further by treating reconstruction errors and SHAP contributions jointly, analyzing how co-occurring anomalous features reinforce each other. For instance, the explanation explicitly notes how SHAP values help to contextualize the anomaly in

[4] https://github.com/Gradiant/Natural-Language-Explanations-4-AD

feature "Fwd Seg Size Min". Moreover, this approach also links the XAI outputs to plausible attacker TTPs, such as Nmap fragmentation scans, producing an explanation that is both technically grounded and operationally informative for cybersecurity analysts.

However, it is worth noting that the explanation module is inherently dependent on the autoencoder's output. Consequently, if the autoencoder misclassified a benign flow as anomalous (false positive), the explanation module does not autonomously correct this prediction. Instead, it attempts to contextualize the classification by explaining the specific feature deviations which triggered the detector. This transparency also serves as a reliability mechanism for the human analyst by exposing the specific logic behind a false positive (e.g., highlighting that a minor deviation in packet size was the sole driver of an "anomaly"). Additionally, the controlled incorporation of alert and explanation pairs of resolved false positives into the retrieval knowledge base would assist the model to identify and flag potential misclassifications. However, this introduction must be carefully limited to prevent the model from misclassifying subtle attacks as false positives.

Our evaluation results demonstrate that while larger models such as Qwen2.5-32B generally achieve higher scores, smaller models can deliver competitive performance with a significant reduction on resource consumption when enhanced with contextual retrieval strategies. In fact, smaller models like Mistral-7B and Gemma-9B with HITL, were able to outperform the explanations produced by larger models like Qwen2.5-32B lacking this grounding. These findings show that the effective use of context strategies and the integration of human knowledge can have a greater impact than the model size alone. This is particularly significant for organizations seeking to balance the need for high-quality, human-aligned explanations with the practical constraints of hardware resources and data privacy.

From an end-to-end perspective, the total time from alert to explanation is dominated by the feature attribution and NLE generation stages. While the initial anomaly detection by the autoencoder is fast ($2794 \pm 89\ \mu s$), the SHAP analysis already introduces a notable delay, with an average runtime of 19.83 ± 0.90 seconds per flow. This is caused by the evaluations of the model on perturbations of the input to approximate feature contributions, a process that scales poorly with the number of features and background samples. However, the final NLE generation is the most time-consuming phase, with generation times ranging from 28.42 to 87.88 seconds in average for the HITL approach.

In a real-world SOC environment, this approach remains viable for SOC operations, as explanation generation is typically not on the critical path of a real-time threat containment. Anomaly detection and initial response actions (traffic throttling or temporary isolation) are often automated and executed immediately after detection. While total generation times include RAG, input processing, and text generation, user wait time can be reduced by presenting the explanation incrementally as it is being generated. The explanation can therefore be generated asynchronously and delivered on-stream, allowing the analyst to initially review the alert and form preliminary hypotheses, while the explanation can be accessed once text generation begins. In this context, a practical mitigation strategy is to temporarily place the detected anomalous flow in a hold or quarantine state after it is flagged, allowing potential threats to be contained while the analyst reviews both the alert and the generated explanation.

Despite yielding higher-quality outcomes, the RAG strategy introduces two performance bottlenecks that contribute to longer inference times. The primary bottleneck is the computational overhead required to process the expanded context window, as the prompt must incorporate all the retrieved information. A secondary, but still significant, source of delay is the retrieval process itself, which adds latency prior to generation. This illustrates a central trade-off in deploying HC-XAI systems: although RAG produces explanations with the highest degree of human alignment, its latency (especially in real-time or resource constrained settings) can limit practical deployment, even when using optimized models.

The HC-XAI pipeline proposed in this work presents a shift from automated alerting to augmented intelligence in security operations. By translating raw anomaly predictions and SHAP values into an intuitive NLE, grounded with expert knowledge, the proposed HC-XAI system is able to provide immediate situational awareness in complex cybersecurity environments. The integrated chat-bot also empowers analysts to move beyond this initial explanation, allowing them to clarify ambiguities and explore hypotheses in a natural, conversational manner. This fusion of automated summarization and interactive dialogue ensures that explanations are not merely delivered, but are truly understood, fostering a collaborative partnership between the human expert and the AI system.

5 Conclusions

This study contributes to the field of HC-XAI by presenting a pipeline to improve model interpretability and enable more effective cybersecurity operations. We show that combining technical feature attribution techniques like SHAP with Natural Language Explanations creates a more robust and human-centered approach to XAI. By using LLMs to generate these explanations, complex outputs are translated

into interpretable and actionable insights suitable for security analysts and other users who may not be experts in ML.

Furthermore, we analyze different human-in-the-loop strategies, showing how AI systems can be designed to support human expertise. We propose a novel evaluation framework for HC-XAI, establishing a benchmark for multiple instruction-tuned LLMs. With different explanation strategies, this study also provides an empirical analysis for understanding the trade-offs for generating domain-specific explanations of model predictions. The evaluation, conducted using the LLM-as-a-Judge framework, was guided by a detailed rubric specifically designed to reflect the cognitive and specific needs of the end user, in this case, a SOC analyst. The obtained results demonstrate that the integration of the proposed HITL approach enables models to improve the most crucial dimensions for an analyst, such as content accuracy and threat awareness. For example, in the evaluation of the Threat Awareness category, Mistral-7B's score improves from 5.32 in the zero-shot setting to 8.25 with contextual retrieval, a score close to the 8.36 achieved by the far larger Qwen-32B model. Therefore, we show that smaller models are able to have a near-equivalent performance with almost half the computational cost when provided with the right HITL approach. This emphasizes the core conclusion of this work: the use of context and integration of human knowledge have a greater impact than model size for developing more practical and accessible NLE-based XAI systems.

Overall, this study highlights the importance of designing scalable and explainable AI systems that are not only technically robust but are fundamentally aligned with human cognitive and domain-specific requirements. By refining AI-generated explanations to align with expert knowledge, this approach supports more trustworthy and accountable AI-driven decision-making in critical contexts. Ultimately, this work promotes the responsible adoption of AI in critical fields like cybersecurity, ensuring that AI serves as a trusted advisor, empowering users to make final decisions with confidence and accountability.

5.1 Limitations and Future Work

Future work aims to create a formal user-interaction study with cybersecurity experts to validate the entire pipeline, including the currently unevaluated interactive chat-bot. This study is essential to move beyond the limitations of the LLM-as-a-Judge approach, allowing us to measure real-world adaptation and impact on analyst trust, decision confidence, and mean time to resolution improvement. Beyond this crucial validation, several promising research avenues emerge from these findings.

First, we aim to explore dynamic explanation personalization by modeling individual analyst expertise and cognitive preferences over time. In this way, the system could learn to adapt the verbosity, technical depth, and format of the NLEs to each analyst. Second, we plan to compare the effectiveness of our approach against more traditional, resource-intensive HITL methods, such as Reinforcement Learning from Human Feedback, to evaluate the trade-offs between these approaches in terms of analyst cognitive load and the long-term impact on human-AI collaboration. Third, we intend to test emerging local open source models, such as gpt-oss (OpenAI, 2025), focusing on their usability, practicability, and security implications in real-world operational contexts. Third, we plan to examine the integration of this HC-XAI pipeline into broader security platforms, analyzing how HC-XAI can move beyond a standalone tool to become an embedded component that enhances human agency and accountability across the organizational security posture. Finally, we will explore extending the feedback loop to enable analyst input not only to refine explanations but also to inform retraining or adaptation of the underlying autoencoder, thereby investigating how human feedback could improve both interpretability and detection performance over time.

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Data Availability The benchmark dataset analyzed in this study is the CSE–CIC-IDS2018 maintained by the Canadian Institute for Cybersecurity, University of New Brunswick, openly accessible at https://www.unb.ca/cic/datasets/ids-2018.html. The evaluation data used to assess the models and the various data generation strategies consist of the predictions produced by the autoencoder in combination with SHAP outputs, applied to the attack instances within the dataset. This evaluation data, as well as prompt details, are available at GitHub (https://github.com/Gradiant/Natural-Language-Explanations-4-AD).

Declarations

Competing interests V. Carneiro reports financial support was provided by Spain Ministry of Science and Innovation. V. Carneiro reports financial support was provided by Government of Galicia. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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